Floquet Cooling

Goal

Can we develop a theory of entropy and energy transport using generic many-body Floquet modulated hamiltonians, that is, for example modulated hamiltonians of the form

$$H_{\mathrm{M}}(t) = \sum_{\boldsymbol{x}} f(\boldsymbol{k} \cdot \boldsymbol{x} - \omega t) h_{\boldsymbol{x}}$$

where $f(\theta)$ is some periodic function, and $h_{\boldsymbol{x}}$ is some local hamiltonian term.

In particular, can a many body system H_{S} be cooled to lower entropy by attaching such a system as an ancilla

$$H = H_{\mathrm{S}} + H_{\mathrm{M}} + H_{\mathrm{int}}$$

High level motivating questions

- Under what circumstances does H_{M} set up an energy/entropy current?
- Are there any general bounds or no-go theorems? (eg. initial calculations in a toy model indicate the heating and energy current are fixed together)
- Are there realistic models that may be useful for cooling many body systems?

Background

- To study quantum effects, or implement quantum algorithms, or make quantum codes, we typically have to make systems very cold, typically close to their ground state energy density
- In the context of cold atom arrays, this is achieved typically by one of two strategies
  1. Tweezer trapping + ramp
     a. Individually trapping atoms in tweezers, which are approximately decoupled
     b. Performing a measurement feedback protocol until all/almost all of the atoms are in the lowest energy state of their tweezer potentials. This produces a system close to its global ground state (but not the ground state of the desired hamiltonian)
     c. Quasi-adiabatically ramp into the target hamiltonian. This must be done sufficiently fast that the system does not decohere, and so the system inevitably heats.
  2. BEC + ramp
     a. Laser/doppler cool a large number of atoms

b. Trap them in a magneto-optical trap

c. Evaporatively cool them (preferentially allow excited atoms to escape) until a BEC forms. This produces a system close to its global ground state (but again, not the ground state of the desired hamiltonian)

d. Quasi-adiabatically ramp the BEC the target hamiltonian. This usually involves turning on a lattice. This must be done sufficiently fast that the system does not decohere, and so the system inevitably heats.

  - For more info see Wieman, C. E., Pritchard, D. E., & Wineland, D. J. (1999). Atom cooling, trapping, and quantum manipulation. *Reviews of Modern Physics*, *71*(2), S253.

    > Atom cooling, trapping, and quantum manipulation
    >
    > Neutral atoms and ions can now be cooled, trapped, and precisely manipulated using laser beams and other electromagnetic fields. This has made it possible to control external degrees of freedom at the quantum level, thereby making possible a variety of interesting new physics.
    >
    > https://doi.org/10.1103/RevModPhys.71.S253

- These techniques cool systems to phenomenally low temperatures in absolute units, (~nanokelvin, i.e. ~1billion times colder than the cosmic microwave background), however the natural energy scales of the system are also very low (~kHz), consequently these are generally not particularly low temperatures in the natural units of the system.

$$\frac{\hbar(1\mathrm{kHz})}{k_{\mathrm{B}}(10^{-9}\mathrm{K})} \approx 8$$

  - Notably here, in De Marco et al, they cool a gas of fermionic polar molecules to $\sim 100\mathrm{nK}$, but, measured in units of the Fermi temperature, this only gives $T/T_{\mathrm{F}} = 0.3$, barely low enough to see the effects of quantum degeneracy/

    > A degenerate Fermi gas of polar molecules
    >
    > A bulk gas of fermionic potassium-rubidium molecules is cooled down to below a third of the Fermi temperature.
    >
    > https://doi.org/10.1126/science.aau7230

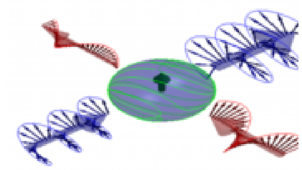
- Consequently there is a sub-literature on various kinds of manipulations to cool quantum simulators to lower temperature. These proposals usually involve either controlled dynamics, measurement feedback or both.

- A sub-group of quantum dynamical effects which can be used for quantum cooling involve energy/entropy pumping (depending on your perspective)

  - A well known group of these (which we will *not* pursue) involve topological effects based on variants of the Thouless pump

- Martin, I., Refael, G., & Halperin, B. (2017). Topological frequency conversion in strongly driven quantum systems. *Physical Review X*, *7*(4), 041008.

> Topological Frequency Conversion in Strongly Driven Quantum Systems
> Spatial dimensionality of a system determines the types of phenomena it can exhibit. A new analysis shows how the dimensionality of a single spin-1/2 particle can be effectively increased by applying a drive with two incommensurate
> https://doi.org/10.1103/PhysRevX.7.041008



- Long, D. M., Crowley, P. J., & Chandran, A. (2021). Nonadiabatic topological energy pumps with quasiperiodic driving. *Physical Review Letters*, *126*(10), 106805.

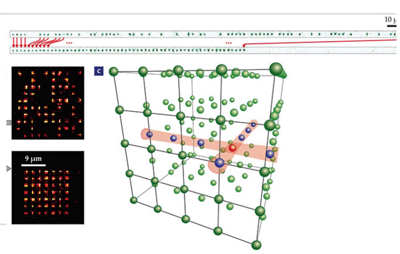
> Nonadiabatic Topological Energy Pumps with Quasiperiodic Driving
> We derive a topological classification of the steady states of d-dimensional lattice models driven by D incommensurate tones. Mapping to a unifying ($d+D$)-dimensional localized model in frequency space reveals anomalous localized topological phases (ALTPs) with no static analog. While the formal classification is
> https://doi.org/10.1103/PhysRevLett.126.106805

- However, another interesting proposal involves simply spatially deforming the hamiltonian. The only existing work (that I know of) thinking in these terms is from Ashvin's group, where they specifically focus on spatially modulated conformal field theories
  - Wen, X., Fan, R., & Vishwanath, A. (2022). Floquet's refrigerator: Conformal cooling in driven quantum critical systems. *arXiv preprint arXiv:2211.00040*.

> Floquet's Refrigerator: Conformal Cooling in Driven Quantum...
> We propose a general method of cooling -- periodic driving generated by spatially deformed Hamiltonians -- and study it in general one-dimensional quantum critical systems described by a conformal...
> https://doi.org/10.48550/arXiv.2211.00040

  - The Vishwanath proposal is very natural to implement in Rydberg-tweezer experiments, where the strength of the local Hamiltonian can be easily controlled. Specifically, the Hamiltonian consists of two-body interactions which decay with inter-particle spacing, and positions of particles can be easily individually controlled. Some simple overviews of Rydberg systems are below:
    - A high level introduction which focuses on applications in quantum computing

> Quantum computing with neutral atoms
> With their hyperfine states serving as two-level qubits, atoms can be packed into closely spaced, laser-cooled arrays and be individually addressed using laser pulses.
> PT https://physicstoday.aip.org/features/quantum-computing-with-neutral-atoms?utm_source=chatgpt.com

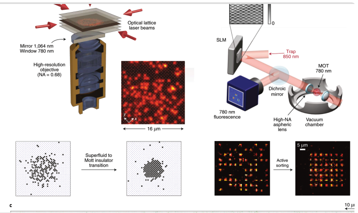


- A more detailed introduction, which focusses on Many Body Physics

Many-body physics with individually controlled Rydberg atoms

Nature Physics - This Review Article outlines the techniques necessary for the manipulation of neutral atoms and making use of their interactions, when excited to Rydberg states, to achieve the...

https://doi.org/10.1038/s41567-019-0733-z



- We are motivated to understand the physics of space-time modulated many body systems with a focus on

Technical goals

Characterise single particle transport: a simple place to start is with a non interacting system. In this setting it is most natural to analyse the particle (rather than energy/entropy) current.

- **Classical hopping problems**: A simple diffusive model consists of a random walk on a line, where the particle, with position hops left or right with rate $\gamma(kx - \omega t)$. A simple argument suggests that in this model the particle develops a drift in the opposite direction to the natural velocity $v = \omega / k$.
- **Ballistic single quantum particle**: A natural next step is to similarly characterise the quantum problem. The same intuition as above applies, however coherence effects may change the physics. This model will teach us about the physics of a generic interacting problem in the regime $\lambda k \gg 1$ where λ is the scattering length

$$H = -t \sum_x f(kx - \omega t) \left| x + 1 \right\rangle \left\langle x \right| + \text{h.c.}$$

- **Diffusive single quantum particle**: This model ties together the above two models, and may be modelled using a Liouvillian model, in which we combine the Hamiltonian dynamics above, with a dissipative term which localises the particle (i.e. suppresses off-diagonal elements in the density matrix ρ).
- **Non interacting many body models**: it should be straightforward to extend the previous analyses to Free fermions, which are fundamentally single particle in nature, with Pauli statistics being the only added ingredient.

Move to solvable many body models: a natural next step is to consider energy current in exactly solvable many body models. The transverse field Ising model is a good start: this model maps to free fermions, but without particle number conservation, so we can study energy rather than particle transport.

Finally study generic interacting models: here we can use what we learned in the previous sections, and lean on structure from ETH (eigenstate thermalisation hypothesis), FGR (Fermi golden rule) and statistical mechanics for additional structure.